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TGTrack: Temporal Generative Learning for Unified Single Object Tracking

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Abstract

Existing single object trackers typically treat temporal modeling superficially by passing limited inter-frame information, such as propagated tokens or template updates, without intrinsic temporal supervision learning. To address this limitation, we propose TGTrack, a new unified tracking framework that incorporates a temporally generative supervision task to guide the model in learning temporal dynamics. The core of TGTrack is a temporally generative learning paradigm equipped with a transformer-based generative decoder, which consists of a gated fusion module and an autoregressive prediction mechanism. This joint design enables the model to infer future scenarios from preceding information, thereby improving its ability to model both visual appearance and temporal dynamics. Furthermore, we introduce a time token embedding to explicitly encode the temporal position of each frame. Experiments on 11 benchmarks spanning five modalities show that TGTrack achieves state-of-the-art performance in robust unified tracking. For instance, TGTrack-B384 achieves an AUC of 75.3% on LaSOT. Code is available at https://github.com/wtg1/TGTrack.

1. Introduction

Single object tracking (SOT) is a fundamental task in computer vision, which aims to continuously locate a specific target throughout a video sequence given its initial bounding box. With the rapid advancement of deep neural networks and large-scale training, recent RGB-based trackers [1, 8, 36, 68, 69] have achieved remarkable performance. To handle more challenging scenarios such as low illumination, severe occlusion, or fast motion, researchers have extended SOT to multi-modal tracking by incorporating complementary modalities such as depth [61], thermal infrared [26, 27], event streams [52], and natural language descriptions [31, 51]. Building upon these advances, recent works [11, 20, 21, 54, 72] have aimed to develop unified frameworks capable of jointly modeling multiple modalities, showing promising results in tracking robustness.

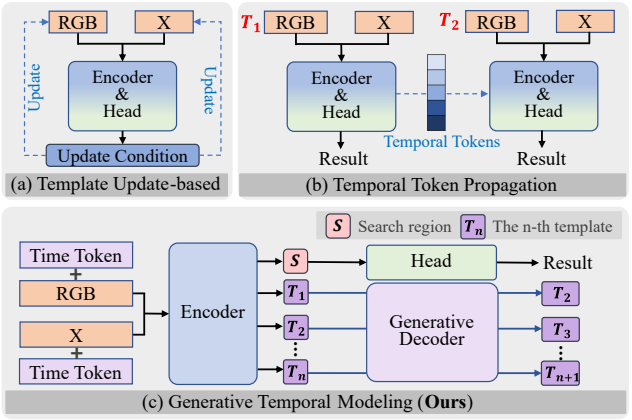


Figure 1. Comparison of three temporal modeling paradigms in tracking: (a) template update-based, (b) token propagation-based, and (c) our generative temporal modeling approach.

However, both RGB-based and unified trackers generally tend to overemphasize architectural design and modality fusion, while overlooking the importance of temporal modeling, which limits their adaptability to dynamic target appearance changes. To address this issue, recent works have explored temporal modeling strategies that can be broadly categorized into two paradigms. The first paradigm, shown in Fig.1(a), updates template frames with handcrafted parameters [9, 11, 23, 59], making the model sensitive to parameter settings and limiting its generalization. The second paradigm, shown in Fig.1(b), propagates a small number of temporal context tokens across frames [1, 22, 28, 57, 68, 70] to capture temporal dynamics. Despite these advances, both paradigms still encode temporal information implicitly by injecting temporally relevant contents into the input but lack explicit supervision, hindering their ability to model temporal dependencies and appearance transitions.

Motivated by the aforementioned limitation, we seek to reformulate tracking supervision by introducing a time-aware learning objective that explicitly guides the model to understand how targets and scenes evolve over time, thereby

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enhancing tracking performance. To this end, we propose TGTrack, a temporal generative unified tracking framework, as illustrated in Fig. 1(c). Unlike previous trackers that rely solely on spatial supervision such as template matching, TGTrack introduces a parallel temporal generative task that encourages the model to actively explore and internalize temporal evolution, transforming temporal modeling from passive reception into active understanding.

Specifically, TGTrack employs a transformer-based temporal generative decoder, which autoregressively takes the accumulated historical frame information as input and predicts the evolution of the target and scene in the next frame. This enhanced supervision through temporal prediction enables a more comprehensive and deeper understanding of temporal dynamics. To support this sequential generation process, we further design a novel time token embedding, which explicitly encodes the temporal position of each frame and guides the model to reason over time in a structured and interpretable manner. Together, these two temporal-aware designs transform temporal modeling from a passive and implicit supplementation into an active and explicit learning process for predicting temporal variations, thereby empowering TGTrack to capture motion continuity and appearance transitions more effectively.

Experimental results show that TGTrack achieves state-of-the-art performance on five tracking tasks across 11 benchmarks. For example, TGTrack-L256 attains an AUC of 75.8% on LaSOT, surpassing the recent unified tracker SUTrack-L224 [11] by 2.3%. Notably, TGTrack-L384 further achieves unprecedented performance on RGB-D/T/E/L benchmarks, with 67.5% F-score on DepthTrack and AUCs of 63.3%, 65.7%, and 68.6% on LasHeR, VisEvent, and TNL2K, respectively. These results highlight the strong generalization of our temporal generative learning. In summary, our main contributions are as follows:

- We propose TGTrack, a novel tracking framework that introduces a unified temporal learning paradigm for five representative tracking tasks.
- We introduce two key technical components for the temporal generative learning framework, including a temporal generative decoder that enables predictive temporal modeling, and a time token embedding that explicitly encodes temporal positions.
- We develop a family of TGTrack models, which achieve state-of-the-art performance across five multi-modal tracking tasks.

2. Related Work

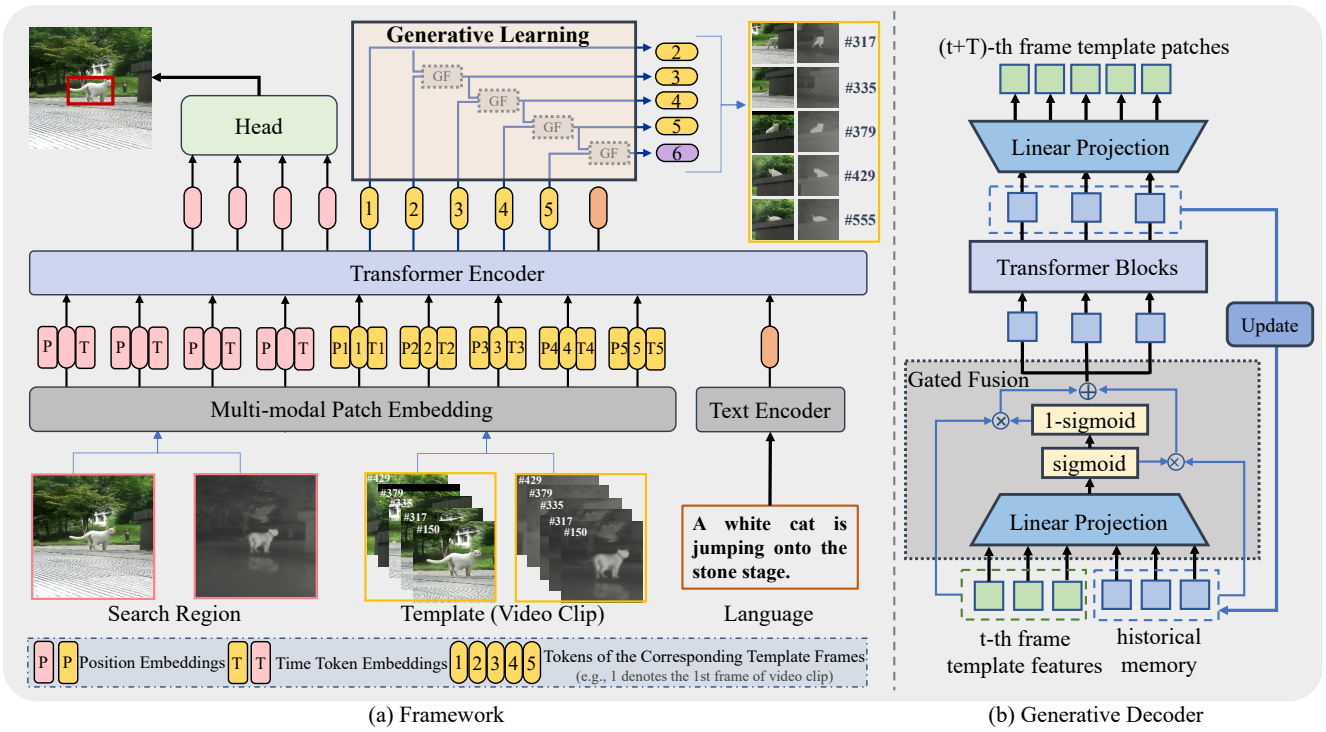
Unified Single Object Tracking. Unified single object tracking is the task of leveraging multiple input modalities, such as RGB, depth, thermal, event data, and language descriptions, to enhance the single object tracking process. Compared with RGB-based tracking [5, 6, 8, 12, 18, 37, 47, 53, 55, 59, 65, 66, 68], multi-modal tracking benefits from the complementary information provided by additional modalities, thereby enhancing robustness and improving performance under challenging conditions. Recent research has increasingly focused on developing unified tracking frameworks that support multiple modalities within a single model. ViPT [72], SDSTrack [21], OneTracker [20], STTrack [22], SeqTrackv2 [10] employ unified frameworks by finetuning a pretrained RGB tracker [62], but still require separate training for each modality, and most of them only support a subset of modalities. Un-Track [54] takes a further step toward model-level unification across these tasks, yet its modality coverage remains limited, excluding RGB-only and RGB-Language tracking. SUTrack [11] achieves the first comprehensive unification across all five multi-modal tracking tasks within a single model.

However, both existing RGB-based and unified trackers still exhibit limited exploration of temporal modeling, which is crucial for capturing dynamic target and scene variations in tracking. Given the broader scope of unified tracking, we propose a new temporally generative tracking framework built upon the unified setting to more comprehensively validate our temporal modeling approach.

Temporal Modeling in Tracking. To overcome the limitations of relying solely on the initial template, tracking frameworks that incorporate temporal information to capture target motion and appearance variations have gradually emerged. Some methods rely on simple template update mechanisms [11, 23, 39, 59], while others propagate a limited number of temporal tokens across frames [22, 46, 53, 56, 68]. However, these approaches typically treat temporal cues as auxiliary information, lacking explicit temporal supervision signals and thus remaining insufficient for modeling complex temporal variations. Different from previous works, we approach this problem from a generative learning perspective and propose a novel temporal learning framework that explicitly supervises the model to learn temporal-aware representations by predicting future frames, thereby enhancing the overall temporal tracking performance.

Generative Learning. Recent progress in generative models has demonstrated remarkable performance in tasks like image synthesis and representation learning. Methods like DVAEs [43] focus on learning the underlying distribution of input data to generate high-quality samples. GAN-based models [49, 58, 74] leverage adversarial training to generate realistic and diverse content, while diffusion models [19, 45] iteratively refine noisy inputs to produce high-quality images. These models primarily aim to enhance the fidelity and diversity of generated content.

In contrast, our work shifts the focus from high-quality feature generation to learning temporal dynamics through a generative perspective. Supervised by continuous future-frame prediction, our method encourages a more proac-



(a) Framework (b) Generative Decoder

Figure 2. **(a) Overall framework of TGTrack.** TGTrack encodes the search region, a clip of video frames, and language description as tokens via a transformer encoder. Search-related features are used for target prediction, while video clip-related features are fed into a generative decoder for temporal modeling. **(b) Architecture of the generative decoder.** The transformer-based generative decoder with gated fusion autoregressively generates future target representations based on accumulated historical frames, enabling temporal reasoning.

tive understanding of temporal evolution. Although ARTrackV2 [1] also employs a generative objective, it is limited to reconstructing target appearance for template updates. Our approach instead uses generation as a temporal learning signal to model frame-to-frame transitions, making it fundamentally different.

3. TGTrack

In this section, we provide a comprehensive description of TGTrack, a temporally generative unified tracking framework. We first present the overall framework, followed by detailed explanations of the two key components, temporal generative decoder and time token embedding. Finally, we depict the prediction head and loss function.

3.1. Overall Framework

The overall framework of TGTrack is depicted in Fig. 2(a). It adopts a unified one-stream transformer architecture, taking a search region $\mathbf{S} \in \mathbb{R}^{6\times H_x\times H_x}$, a video clip (serving as references) $\mathbf{V} \in \mathbb{R}^{L\times 6\times H_z\times H_z}$, and a language description as input. Inspired by SUTrack [11], both the search region and the video clip are formed by channel-wise concatenation of RGB and other modality images, enabling the model to handle various modality combinations. First, $\mathbf{S}$ and $\mathbf{V}$ are converted into patch embeddings with a stride of 16, resulting in $\mathbf{P_s} \in \mathbb{R}^{C\times \frac{H_x}{16}\times \frac{H_x}{16}}$ and $\mathbf{P_v} \in \mathbb{R}^{L\times C\times \frac{H_z}{16}\times \frac{H_z}{16}}$, respectively. Meanwhile, the language description is processed by a text encoder composed of CLIP[42] and a linear projection layer, producing a text embedding $\mathbf{T} \in \mathbb{R}^{C\times 1}$. Next, positional embeddings and time token embeddings are added to $\mathbf{P_s}$ and $\mathbf{P_v}$ to incorporate spatial and temporal information, respectively. Then, $\mathbf{P}_s$, $\mathbf{P}_v$ and $\mathbf{T}$ are flattened along the spatial dimensions and concatenated into a single sequence $\mathbf{P_{svt}} \in \mathbb{R}^{C\times L_{svt}}$ ($L_{svt} = \frac{H_x}{16}\times\frac{H_x}{16} + L\times\frac{H_z}{16}\times\frac{H_z}{16} + 1$), which is fed into a transformer encoder to extract multi-modal features. Among the extracted features, those corresponding to the search region are passed to a tracking prediction head for target localization, while those associated with the video clip are fed into a generative decoder. The decoder predicts future frames through temporal generative learning, enhancing model's ability to exploit temporal information.

3.2. Temporal Generative Learning

Temporal generative learning is designed as an autoregressive process. Among the features extracted by the transformer encoder, those corresponding to the video clip are used as input to this generative process. The video clip con-

tains L frames sampled from the original sequence, preserving sparse but long-range temporal correlations. The process is conducted sequentially according to the frame order in the video clip. At each step, it generates representations of the next frame based on the current frame and all preceding frames. Specifically, the second frame is generated from the first, the third from the first and second, and so forth. At the l-th step, the model predicts the $(l+1)$-th frame representations conditioned on the current and all previous $l-1$ frame features. Through temporal generative learning, TG-Track is able to effectively model temporal dynamics across frames, which in turn enhances tracking performance.

The generative decoder is the core of temporal generative learning. It is designed to autoregressively model temporal dependencies across frames. The decoder is built upon a stack of transformer blocks and integrates a gated fusion mechanism to dynamically combine historical context with the current input, as shown in Fig. 2(b). Its input comprises features extracted by the transformer encoder that correspond to the given video clip. Formally, we use $\mathbf{F} \in \mathbb{R}^{L \times \frac{H_z}{16} \times \frac{H_z}{16} \times C}$ to denote the extracted feature tensor, where L is the sequence length of the video clip, and C is the embedding dimension. The feature corresponding to the t-th frame is denoted as $\mathbf{F}_t$.

The generative decoding process begins with an initial historical memory $\mathbf{H}_0 = \mathbf{0}$. At each time step $t \in 1, 2, \ldots, L$, the current frame features $\mathbf{F}_t$ and the memory from the previous step $\mathbf{H}_{t-1}$ are concatenated along the spatial dimensions to form Z_t, as follows:

$$\mathbf{Z}_t = \mathrm{concat}(\mathbf{H}_{t-1}, \mathbf{F}_t). \tag{1}$$

The concatenated features are then passed through a gated fusion module, which dynamically adjusts the contributions of historical memory and current features through a sigmoid-based gating function. This design allows the model to effectively preserve and utilize semantic information from early frames (e.g., t=1) when predicting future ones (e.g., t+n), thereby alleviating the feature degradation issue commonly observed in traditional methods due to limited temporal modeling scope. Formally, the gating vector at time step t is computed as:

$$\mathbf{G}_t = \sigma(\mathbf{W}_g \mathbf{Z}_t + \mathbf{b}_g), \tag{2}$$

where $\mathbf{W}_g$ is a learnable weight matrix, $\mathbf{b}_g$ is the bias term, and $\sigma(\cdot)$ denotes the element-wise sigmoid activation function. The gating vector $\mathbf{G}_t$ is then used to adaptively fuse the current input feature $\mathbf{F}_t$ with the previous memory $\mathbf{H}_{t-1}$, summarized as follows:

$$\tilde{\mathbf{F}}_t = \mathbf{G}_t \odot \mathbf{H}_{t-1} + (1 - \mathbf{G}_t) \odot \mathbf{F}_t, \tag{3}$$

where $\odot$ denotes element-wise multiplication. The fused feature $\tilde{\mathbf{F}}_t$ is subsequently refined through a stack of transformer blocks to enhance its spatiotemporal representation:

$$\mathbf{F}_t^{\text{out}} = \mathrm{TransformerBlocks}(\tilde{\mathbf{F}}_t), \tag{4}$$

which is also used to update the historical memory for the next step:

$$\mathbf{H}_t = \mathbf{F}_t^{\text{out}}. \tag{5}$$

To predict the representation of the next frame, the refined output is first normalized and then projected back to patch space via a linear prediction head:

$$\hat{\mathbf{V}}_{t+1} = \mathbf{W}_p \cdot \mathrm{LN}(\mathbf{F}_t^{\text{out}}) + \mathbf{b}_p, \tag{6}$$

where $\mathbf{W}_p$ and $\mathbf{b}_p$ represent the weight matrix and bias term of the linear prediction head, respectively, and $\mathrm{LN}(\cdot)$ denotes the layer normalization operation. $\hat{\mathbf{V}}_{t+1}$ serves as the predicted representation of t+1-th frame, and is supervised by the ground-truth via a mean squared error (MSE) loss:

$$\mathcal{L}_{\text{gen}} = \frac{1}{L} \sum_{t=0}^{L} \left\| \hat{\mathbf{V}}_{t+1} - \mathbf{V}_{t+1} \right\|_2^2, \tag{7}$$

where $\mathbf{V}_{t+1}$ represents the ground-truth t+1-th frame representation, which is transformed into patch embeddings through a patchification operation.

3.3. Time Token Embedding

To enhance the temporal awareness of the encoder, we introduce a time token embedding that combines polar-coordinate-based initialization with trainable temporal modulation. While most existing methods treat frame features as unordered sets, we introduce frame-wise temporal encoding. This mechanism provides each frame with a unique temporal identity, enabling the model to distinguish not only "what" is observed, but also "when" it is observed, a capability critical for robust temporal reasoning.

A learnable base time token $\mathbf{t}_0 \in \mathbb{R}^C$, which serves as the initial temporal representation. To differentiate the temporal positions of individual frames, we apply a polar transformation governed by a set of angles $\theta = \{\theta_1, \theta_2, ..., \theta_L\}$, where L is length of the video clip. Each angle determines a unique rotation of the shared time token, defined as:

$$\mathbf{T}_t = \cos(\theta_t) \cdot \mathbf{t}_0 + \sin(\theta_t) \cdot \mathbf{W}(\mathbf{t}_0), \tag{8}$$

where $\theta_t \in [0, \frac{\pi}{2}]$ denotes the angle of the t-th frame, and $\mathbf{W}$ is a learnable linear transformation. This formulation allows each frame to receive a distinct learnable temporal encoding while preserving a shared structure across frames.

By default, the angles θ_t are initialized uniformly from 0 to $\frac{\pi}{2}$ to ensure a gradual variation in temporal representations. When the learnable angle mode is enabled, the model is allowed to optimize these angles during training, effectively learning the optimal temporal spacing among frames.

The resulting time token embeddings, denoted as $\{\mathbf{T}_1, ..., \mathbf{T}_L\}$, are broadcasted across the batch dimension and then directly added to the corresponding patch embeddings of each frame, along with the position embeddings. It can be formulated as:

$$\mathbf{E}_t = \mathbf{E}_t^{\text{init}} + \mathbf{T}_t + \mathbf{P}_t, \tag{9}$$

where $\mathbf{E}_t$ represents the final embedding for the t-th frame, $\mathbf{E}_t^{\text{init}}$ denotes the initial patch embedding, $\mathbf{T}_t$ is time token embedding, and $\mathbf{P}_t$ is position embedding. By summing these components, the model encodes both the spatial layout and the temporal order in a unified embedding space.

Thus, the time token embedding provides the encoder with explicit and learnable temporal cues, enabling TGTrack to model temporal dependencies more effectively without significant computational overhead.

3.4. Prediction Head and Loss Function

The prediction for temporal generation has been detailed in Sec. 3.2. For tracking prediction, the features corresponding to the search region, extracted by the transformer encoder, are fed to a tracking prediction head to generate the final tracking results. We adopt the prediction head of OSTrack [62], which includes a classification branch for target presence estimation and a regression branch for bounding box localization. To train the model, we utilize three types of loss functions. First, a focal loss [35] $\mathcal{L}_{\text{cls}}$ is employed to supervise the classification branch, guiding the model to distinguish the target from the background effectively. Second, the regression loss consists of an ℓ_1 loss and a generalized IoU (GIoU) loss [44], which together supervise the accuracy of the predicted bounding box coordinates. Third, the generative loss $\mathcal{L}_{\text{gen}}$ (Eq. 7) is used to supervise the temporal generative learning. The overall training objective is a weighted sum of the three loss terms:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{cls}}\mathcal{L}_{\text{cls}} + \lambda_{\ell_1}\mathcal{L}_{\ell_1} + \lambda_{\text{G}}\mathcal{L}_{\text{GIoU}} + \lambda_{\text{gen}}\mathcal{L}_{\text{gen}}, \tag{10}$$

where λ_{cls}, λ_{ℓ_1}, λ_{G}, and λ_{gen} act as regularization weights to control the relative influence of each loss component. By default, they are set to 1, 5, 2, and 0.1, respectively.

4. Experiments

4.1. Implementation Details

Model Architecture. We construct six TGTrack variants to comprehensively evaluate the generality of our method. Specifically, TGTrack-L256/384 adopt HiViT-L [63], TGTrack-B256/384 use HiViT-B, while TGTrack-S224 and TGTrack-T224 employ HiViT-S and HiViT-T, respectively. All backbones are initialized with Fast-iTPN [50] weights. The generative decoder is implemented as a stack of eight transformer blocks with channel width and attention heads aligned with the backbone.

Table 1. Details of TGTrack model variants.

Model	Params (M)	FLOPs (G)	Speed (*fps*)
TGTrack-T224	$22^{(+85+20)}$	10	108
TGTrack-S224	$35^{(+85+20)}$	16	74
TGTrack-B256	$70^{(+85+34)}$	44	47
TGTrack-B384	$70^{(+85+34)}$	99	19
TGTrack-L256	$247^{(+86+71)}$	148	17
TGTrack-L384	$247^{(+86+71)}$	333	11

For language-guided tracking, we use CLIP-L [42] as the text encoder. Statistics on model size, FLOPs, and inference speed (measured on a single NVIDIA 2080 Ti) are summarized in Tab. 1. Here, "224", "256", and "384" denote search region resolutions of 224×224, 256×256, and 384×384 (with corresponding template resolutions of 112×112, 128×128, and 192×192), respectively. The parameter count includes the base encoder, while superscripts indicate the additional text-encoder and generative-decoder parameters used only during training. All models are implemented in Python 3.11 with PyTorch 2.8.

Training Setup. TGTrack is trained on a diverse collection of tracking datasets, including RGB datasets, LaSOT [15], TrackingNet [40], COCO [34], VastTrack [41], and GOT-10k [24], as well as multi-modal datasets, RGB-Thermal dataset LasHeR [27], RGB-Event dataset VisEvent [52], RGB-Depth dataset DepthTrack [61], and RGB-Language dataset TNL2K [51]. All datasets are jointly utilized during training to enhance cross-modal generalization.

During training, each input sample consists of one search frame and 5 template frames. The template and search images are cropped by enlarging the ground-truth bounding boxes by factors of 2 and 4, respectively. Data augmentation includes horizontal flipping and brightness jittering. The model is trained for 180 epochs on 4 NVIDIA A100 GPUs, with each epoch containing 100,000 training pairs. We adopt AdamW [38] as the optimizer, setting the initial learning rate to 1×10^{-5} for the transformer encoder and 1×10^{-4} for the remaining parameters, while keeping the text encoder frozen. The learning rate is reduced by a factor of 10 after 144 epochs, with a weight decay set to 1×10^{-4}.

Inference. Our temporal generative decoder provides explicit temporal supervision only during training and is disabled during inference, thus introducing no additional computational overhead.

4.2. State-of-the-Art Comparisons

We evaluate TGTrack against the latest state-of-the-art trackers on 11 widely-used tracking benchmarks, covering five representative tracking tasks, including RGB-based, RGB-Depth, RGB-Thermal, RGB-Event, and RGB-Language tracking. *Benchmark details and additional comparisons are provided in the supplementary materials.*

Table 2. State-of-the-art comparisons across four large-scale benchmarks. The numeral in each method name denotes the search region resolution. The top, second, and third best results are highlighted in **red**, blue, and green, respectively.

	Method	Source	LaSOT			LaSOT$_{ext}$			TrackingNet			GOT-10k		
			AUC	P$_{Norm}$	P	AUC	P$_{Norm}$	P	AUC	P$_{Norm}$	P	AO	SR$_{0.5}$	SR$_{0.75}$
Base Models	**TGTrack-B384**	Ours	**75.3**	**84.9**	**83.1**	**54.8**	**65.7**	**61.6**	**87.5**	**91.6**	**87.9**	79.8	88.5	80.9
	TGTrack-B256	Ours	74.6	84.8	82.4	53.2	64.2	60.4	86.2	90.6	85.9	**80.8**	91.1	**81.6**
	TGTrack-S224	Ours	72.9	83.0	79.8	52.4	63.4	59.6	85.3	89.7	84.5	77.2	86.8	77.6
	SUTrack-B384 [11]	AAAI25	74.4	83.9	81.9	52.9	63.6	60.1	86.5	90.7	86.8	79.3	88.0	80.0
	MambaLCT-B384 [28]	AAAI25	73.6	84.1	81.6	53.3	64.8	61.4	85.2	89.8	85.2	76.2	86.7	74.3
	LMTrack-B384 [57]	AAAI25	73.2	83.4	81.0	53.6	64.7	61.5	85.7	89.9	84.7	80.1	**91.5**	79.0
	ODTrack-B384 [68]	AAAI24	73.2	83.2	80.6	52.4	63.9	60.1	85.1	90.1	84.9	77.0	87.9	75.1
	LoRAT-B378 [33]	ECCV24	72.9	81.9	79.1	53.1	64.8	60.6	84.2	88.4	83.0	73.7	82.6	72.9
	ARPTrack-B256 [32]	CVPR25	72.6	81.4	78.5	52.0	62.9	58.7	85.5	90.0	85.3	77.7	87.3	74.3
	ARTrackV2-256 [1]	CVPR24	71.6	80.2	77.2	50.8	61.9	57.7	84.9	89.3	84.5	75.9	85.4	72.7
	AQATrack-256 [56]	CVPR24	71.4	81.9	78.6	51.2	62.2	58.9	83.8	88.6	83.1	73.8	83.2	72.1
	OneTracker-384 [20]	CVPR24	70.5	79.9	76.5	-	-	-	83.7	88.4	82.7	-	-	-
Large Models	**TGTrack-L384**	Ours	**76.4**	**86.2**	**84.7**	**55.9**	**67.2**	**63.8**	**88.0**	**92.1**	**89.2**	**81.8**	**91.1**	**83.8**
	TGTrack-L256	Ours	75.8	86.0	84.2	54.8	65.4	61.3	87.5	91.7	88.1	81.4	90.7	83.7
	SUTrack-L384 [11]	AAAI25	75.2	84.9	83.2	53.6	64.2	60.5	87.7	91.7	88.7	81.5	89.5	83.3
	ARPTrack-L384 [32]	CVPR25	74.2	83.4	81.7	54.2	64.4	61.2	86.6	91.1	87.4	81.5	90.6	80.5
	LoRAT-L224 [33]	ECCV24	74.2	83.6	80.9	52.8	64.7	60.0	85.0	89.5	84.4	75.7	84.9	75.0
	ODTrack-L384 [68]	AAAI24	74.0	84.2	82.3	53.9	65.4	61.7	86.1	91.0	86.7	78.2	87.2	77.3
	ARTrackV2-L384 [1]	CVPR24	73.6	82.8	81.1	53.4	63.7	60.2	86.1	90.4	86.2	79.5	87.8	79.6
	ARTrack-L384 [53]	CVPR23	73.1	82.2	80.3	52.8	62.9	59.7	85.6	89.6	86.0	78.5	87.4	77.8
	SeqTrack-L384 [8]	CVPR23	72.5	81.5	79.3	50.7	61.6	57.5	85.5	89.8	85.8	74.8	81.9	72.2
	MixViT-L384 [14]	TPAMI24	72.4	82.2	80.1	-	-	-	85.4	90.2	85.7	75.7	85.3	75.1
Efficient Models	**TGTrack-T224**	Ours	**71.3**	**81.3**	**77.7**	**51.2**	**62.1**	**57.6**	**84.3**	**88.9**	**82.6**	**76.1**	**86.7**	**78.4**
	SUTrack-T224 [11]	AAAI25	69.6	79.3	75.4	50.2	61.1	57.0	82.7	87.2	80.8	72.7	82.1	70.5
	AsymTrack-B [73]	AAAI25	64.7	73.0	67.8	44.6	-	-	80.0	84.5	77.4	67.7	76.6	61.4
	HiT-Base [25]	ICCV23	64.6	73.3	68.1	-	-	-	80.0	84.4	77.3	64.0	72.1	58.1
	MixformerV2-S [13]	NIPS23	60.6	69.9	60.4	43.6	-	46.2	75.8	81.1	70.4	-	-	-
	HCAT [7]	ECCVW22	59.3	68.7	61.0	-	-	-	76.6	82.6	72.9	65.1	76.5	56.7
	E.T.Track [2]	WACV23	59.1	-	-	-	-	-	75.0	80.3	70.6	-	-	-
	FEAR-L [3]	ECCV22	57.9	-	60.9	-	-	-	-	-	-	64.5	74.6	-
	LightTrack [60]	CVPR21	53.8	-	53.7	-	-	-	72.5	77.8	69.5	61.1	71.0	-

RGB-based Tracking. We evaluate TGTrack on four RGB-based benchmarks: LaSOT [15] and LaSOT$_{ext}$ [16] for long-term tracking, and TrackingNet [40] and GOT-10k [24] for short-term tracking. As shown in Tab. 2, TGTrack-B384 achieves new state-of-the-art AUC scores of 75.3%, 54.8%, and 87.5% on LaSOT, LaSOT$_{ext}$, and TrackingNet, respectively, outperforming the previous unified tracker (SUTrack) and other RGB-specific trackers. Notably, TGTrack-S224 (only 35M parameters) even surpasses state-of-the-art temporal trackers ARTrackV2 (130M) and AQATrack (72M), demonstrating the strong efficiency and effectiveness of our temporally generative learning framework. In the large-model setting, TGTrack-L384 achieves the best results across all four benchmarks, further validating the scalability of our approach. Moreover, although TGTrack-T224 is not specifically designed as an efficient tracker, it can run in real time on edge devices (26 FPS on an Intel i9-14900KF CPU) and still outperforms previous lightweight trackers by a large margin. Benefiting from explicit temporal supervision, TGTrack maintains superior performance under challenging conditions such as full occlusion and fast motion, as illustrated in Fig. 3.

RGB-Depth Tracking. As shown in Tab. 3, on VOT-RGBD22, TGTrack-L384 achieves a new state-of-the-art EAO of 77.8%, TGTrack-B256 even surpasses the larger SUTrack-L384. On DepthTrack, TGTrack-L384 attains the best F-score of 67.5%, exceeding the latest temporal tracker STTrack [22] by 4.2%. These results demonstrate that the proposed temporal generative learning effectively captures geometric–temporal consistency, thereby further enhancing the model's spatial perception capability.

RGB-Thermal Tracking. As shown in Tab. 3, TGTrack-L384 achieves an AUC of 63.3% on LasHeR, exceeding SUTrack-L384 by 1.4%. On RGBT234, TGTrack-

Table 3. State-of-the-art comparisons on RGB-Depth, RGB-Thermal, and RGB-Event tracking. The best three results are highlighted in **red**, blue and green, respectively.

Method	Source	VOT-RGBD22			DepthTrack			LasHeR		RGBT234		VisEvent	
		EAO	Acc.	Rob.	F-score	Re	Pr	AUC	P	MSR	MPR	AUC	P
TGTrack-L384	Ours	**77.8**	83.3	93.2	**67.5**	**67.4**	**67.6**	**63.3**	**79.5**	**71.3**	**95.2**	**65.7**	**83.0**
TGTrack-L256	Ours	77.7	83.2	93.0	66.8	66.6	67.0	62.7	78.7	71.1	94.9	65.1	82.8
TGTrack-B384	Ours	77.5	82.9	93.2	66.0	66.1	65.9	62.8	78.9	69.5	93.1	64.7	81.6
TGTrack-B256	Ours	77.4	82.9	93.0	65.5	65.7	65.3	61.7	77.4	69.9	93.7	63.9	81.4
TGTrack-S224	Ours	76.6	82.2	92.4	63.4	63.3	63.5	59.4	74.5	68.5	91.4	63.1	80.6
TGTrack-T224	Ours	76.0	81.3	91.7	63.3	63.6	63.0	58.2	73.1	67.0	90.0	62.2	80.4
STTrack [22]	AAAI25	77.6	82.5	**93.7**	63.3	63.4	63.2	60.3	76.0	66.7	89.8	61.9	78.6
CSTrack [17]	ICML25	77.4	83.3	92.9	65.8	66.4	65.2	60.8	75.6	70.9	94.0	65.2	82.4
APTrack [23]	TAI25	77.4	82.1	93.4	62.1	61.9	62.3	58.9	74.1	-	-	61.8	78.5
SUTrack-L384 [11]	AAAI25	76.6	**83.5**	92.2	66.4	66.4	66.5	61.9	76.9	70.3	93.7	63.8	80.5
SMSTracker [4]	ICCV25	74.8	82.2	89.7	63.6	63.1	64.1	56.0	70.3	64.5	86.9	60.4	76.3
XTrack-L [48]	ICCV25	74.0	82.8	88.9	64.8	64.3	65.4	58.7	73.1	65.4	87.8	63.3	80.5
SDSTrack [21]	CVPR24	72.8	81.2	88.3	61.9	60.9	61.4	53.1	66.5	62.5	84.8	59.7	76.7
OneTracker [20]	CVPR24	72.7	81.9	87.2	60.9	60.4	60.7	53.8	67.2	64.2	85.7	60.8	76.7
ViPT [72]	CVPR23	72.1	81.5	87.1	59.4	59.6	59.2	52.5	65.1	61.7	83.5	59.2	75.8
Un-Track [54]	CVPR24	71.8	82.0	86.4	61.0	61.0	61.0	51.3	64.6	62.5	84.2	58.9	75.5

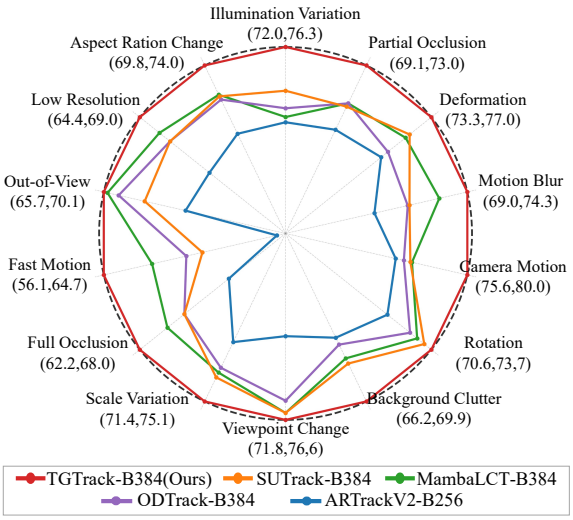


Figure 3. Attribute evaluation of AUC on the LaSOT test set.

L384 further outperforms the latest large-ViT-based tracker XTrack-L [48] by 5.9% in MSR, indicating that the temporal generative mechanism strengthens the utilization of complementary information across modalities.

RGB-Event Tracking. As illustrated in Tab. 3, TGTrack-L384 achieves a new state-of-the-art AUC of 65.7% and Precision of 83.0% on the VisEvent [52] benchmark, effectively capturing motion cues from event information.

RGB-Language Tracking. As presented in Tab. 4, TGTrack-L384 attains an unprecedented AUC of 68.6% on TNL2K, surpassing the recent strongest tracker SUTrack-L384 by 0.7%. On OTB99, TGTrack-L384 outperforms SUTrack-L384 by 1.7% and 3.2% in AUC and Precision, respectively. These results demonstrate the strong semantic alignment capability of our method.

Table 4. SOTA comparisons on RGB-Language tracking.

Method	Source	TNL2K		OTB99	
		AUC	P	AUC	P
TGTrack-L384	Ours	**68.6**	**72.8**	**72.9**	**96.3**
TGTrack-L256	Ours	67.7	71.7	72.5	95.1
TGTrack-B384	Ours	66.7	70.8	70.9	94.6
TGTrack-B256	Ours	65.4	68.7	72.5	95.2
TGTrack-S224	Ours	64.7	67.5	69.9	92.7
TGTrack-T224	Ours	63.0	65.3	69.8	93.3
SUTrack-L384 [11]	AAAI25	67.9	72.1	71.2	93.1
DUTrack-256 [29]	CVPR25	64.9	70.6	70.9	93.9
MMTrack [67]	TCSVT23	58.6	59.4	70.5	91.8
OneTracker [20]	CVPR24	58.0	59.1	69.7	91.5
JointNLT [71]	CVPR23	56.9	58.1	65.3	85.6
TransVLT [64]	PRL23	56.0	-	69.9	91.2
CTRTNL[30]	CVPR22	44.0	45.0	69.0	91.0
TNL2K-II [51]	CVPR21	42.0	42.0	68.0	88.0

4.3. Ablation and Analysis

We conduct ablation experiments based on the TGTrack-B256 across five benchmarks. *More ablation studies are provided in the supplementary materials.*

Effect of model components. As summarized in Tab. 5, we investigate the contributions of the key components in TGTrack. Specifically, removing the generative decoder (#2) leads to an average drop of 1.50%, while excluding the time token embedding (#3) results in a 0.44% decrease. When both components are removed (#4), the performance declines by 2.36%, highlighting the effectiveness of explicit temporal learning for tracking tasks. Removing the gated fusion module (#5) impairs temporal feature learning in the

Table 5. Ablation study on AUC scores (F-score on DepthTrack). 'Δ' denotes the performance change (averaged over benchmarks) compared with the baseline.

#	Variants	LaSOT	DepthTrack	LasHeR	VisEvent	TNL2K	Δ
1	Baseline	74.6	65.5	61.7	63.9	65.4	–
2	W/o GD	73.6	63.5	60.2	62.1	64.2	**-1.50**
3	W/o TTE	74.3	65.1	61.3	63.4	64.8	**-0.44**
4	W/o (2+3)	73.4	61.6	59.0	61.5	63.8	**-2.36**
5	W/o GFM	74.0	64.5	60.9	63.0	64.5	**-0.84**
6	UTTE	74.1	64.6	61.6	62.9	65.0	**-0.58**
7	PCF	73.9	63.9	60.9	63.3	64.6	**-0.90**
8	4 Blocks	74.3	65.2	61.4	63.5	65.1	**-0.32**
	6 Blocks	74.5	65.7	61.0	64.0	65.5	**-0.08**
	10 Blocks	74.5	65.3	61.2	63.8	65.3	**-0.20**
9	3 frames	74.2	65.1	61.3	63.6	65.1	**-0.36**
	4 frames	74.7	65.0	61.8	63.9	65.2	**-0.10**
	6 frames	74.8	65.3	61.7	63.7	65.3	**-0.06**

generative decoder, leading to a 0.84% degradation. Moreover, using a non-learnable time token (#6) further reduces performance by 0.58%, demonstrating the importance of learnable temporal encoding. We also compare different generation targets: predicting the current frame (#7) instead of the next one results in a 0.90% drop, confirming the necessity of the next-frame prediction strategy in capturing temporal dynamics.

Effect of generative decoder depth. We further analyze how the depth of the generative decoder influences performance, as shown in Tab. 5(#8). The default layer depth is set to 8, achieves the trade-off between temporal capability and training cost. Reducing it to a shallower depth leads to insufficient modeling of temporal dynamics, while adopting a deeper decoder increases optimization difficulty, both resulting in degraded performance.

Effect of the video clip length. The sequence length also plays a crucial role. As shown in Tab. 5(#9), shorter sequences (3–4 frames) provide insufficient temporal context and thus degrade tracking performance, while longer ones (6 frames) increase redundancy and tend to distract the model from the primary tracking objective. Using 5 frames achieves the best trade-off between temporal coverage and training stability.

Effect of generative loss weight. The generative loss is designed as a regularizer to learn temporal dynamics without dominating the tracking objective. As shown in Tab. 6, A smaller weight (0.01) weakens temporal supervision, while a larger one (0.5 or 1) makes the model overemphasize generative learning at the expense of tracking accuracy. Thus, setting the weight to 0.1 provides an effective balance between the two objectives.

Visualization. As illustrated in Fig. 4, benefiting from our temporal generative learning mechanism, the generative decoder can accurately predict the target's future appearance and motion patterns based on historical frames, even under challenging conditions such as significant appearance variations (row 1&2) or subtitle occlusion (row 3&4). These observations indicate that our generative decoder goes beyond simple aggregation of historical information. It can effectively predict how the target's appearance evolves in future frames, thereby enhancing the robustness and adaptability of tracking. *More visualizations are provided in the supplementary materials*.

Table 6. Ablation study on the generative loss weight λ_{gen}.

λ_{gen}	LaSOT	DepthTrack	LasHeR	VisEvent	TNL2K	Δ
0.1	74.6	65.5	61.7	63.9	65.4	–
0.01	74.5	65.6	61.5	63.8	65.3	**-0.08**
0.5	74.4	65.4	61.6	64.1	65.3	**-0.06**
1.0	74.2	65.3	61.7	63.7	65.2	**-0.20**

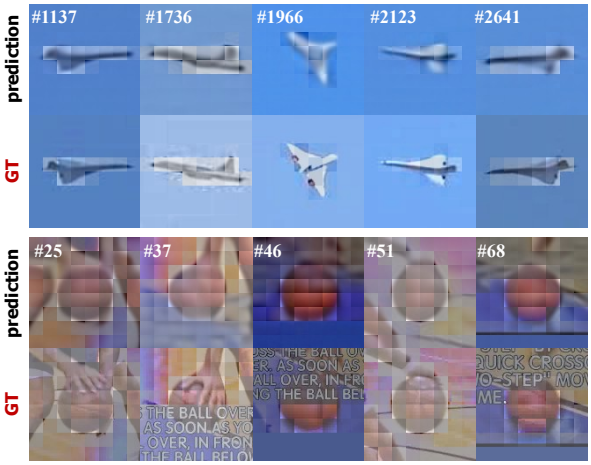


Figure 4. Visualization of generative decoder predictions and ground truths on LaSOT (row1&2) and GOT-10k (row3&4).

5. Conclusion

We propose TGTrack, a unified multi-modal tracking framework that effectively captures temporal information. TGTrack introduces a new temporal generative learning method for tracking, enabling proactive temporal modeling. Extensive experiments across 11 benchmarks spanning five modalities demonstrate that TGTrack consistently achieves state-of-the-art performance, significantly enhancing tracking performance across diverse scenarios.

Limitations. While TGTrack performs well on existing SOT benchmarks, its generalizability to dynamic real-world conditions remains unverified. The current design assumes fixed RGB+X inputs, whereas practical scenarios often feature changing sensor availability or missing modalities. Adapting temporal generative learning to such dynamic settings is an important avenue for future work.

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